

NOTE GENIE

Academic Resource Handbook

Programs · Universities · Courses · Assignments · Books & Learning Resources

Sample Edition v1

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Inside this handbook

Programs · Universities · Semester courses (code, credits, description) · Assignments (Easy/Medium/Hard)
· Recommended books (Free & Paid) · Notes, question papers, lab manuals & project ideas.

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Bachelor of Computer Applications (BCA)

Program code: BCA

Universities offering this program

- IGNOU
- Delhi University (SOL)
- Amity University
- LPU
- Sikkim Manipal University
- Annamalai University

Semester 1

BCA-101 — Programming in C (4 credits)

Fundamentals of C: data types, control flow, functions, pointers, arrays, structures and file handling.

Assignments

1. Write a C program to check whether a number is prime. [\[Easy\]](#)
2. Implement a singly linked list with insert, delete and traverse operations. [\[Medium\]](#)
3. Build a mini library management system using structures and file I/O. [\[Hard\]](#)

Recommended books

- The C Programming Language — Kernighan & Ritchie (Pearson) [Paid](#)
- C Programming (NPTEL) — IIT Kharagpur (NPTEL) [Free](#) · nptel.ac.in
- Beej's Guide to C — Brian Hall (Open) [Free](#) · beej.us

Additional resources

- Lecture notes (Units 1-5)
- Previous-year question paper (2024)
- C Lab manual
- Practice question bank (60 Qs)

BCA-102 — Digital Logic & Computer Fundamentals (4 credits)

Number systems, Boolean algebra, logic gates, combinational & sequential circuits, memory basics.

Assignments

1. Simplify a 4-variable Boolean expression using a K-map. [\[Easy\]](#)
2. Design a 4-bit binary adder and explain carry propagation. [\[Medium\]](#)
3. Design a 4-bit synchronous up/down counter with state diagram. [\[Hard\]](#)

Recommended books

- Digital Design — M. Morris Mano (Pearson) [Paid](#)
- Digital Logic Design (NPTEL) — IIT (NPTEL) [Free](#) · nptel.ac.in

Additional resources

- Unit-wise notes
- Truth-table worksheet
- Previous-year paper (2023)

Semester 2

BCA-201 — Data Structures (4 credits)

Arrays, stacks, queues, linked lists, trees, graphs, sorting & searching with complexity analysis.

Assignments

1. Implement stack-based infix to postfix conversion. [\[Easy\]](#)
2. Implement a binary search tree with insert/delete/search. [\[Medium\]](#)
3. Implement Dijkstra's shortest path on a weighted graph. [\[Hard\]](#)

Recommended books

- Data Structures Using C — Reema Thareja (Oxford) [Paid](#)
- Open Data Structures — Pat Morin (Open) [Free](#) · opendatastructures.org

Additional resources

- DSA notes
- Complexity cheat-sheet
- Lab manual

- 100 practice problems

B.Tech in Computer Science & Engineering

Program code: BTECH-CSE

Universities offering this program

- IIT Delhi
- NIT Trichy
- VIT
- BITS Pilani
- Anna University
- Amity University
- SRM

Semester 3

CS301 — Database Management Systems (4 credits)

Relational model, SQL, ER design, normalization, transactions, indexing and NoSQL overview.

Assignments

1. Write SQL queries for a given schema (joins, group by, sub-queries). [\[Easy\]](#)
2. Normalize a relation up to BCNF and justify each step. [\[Medium\]](#)
3. Design and implement a transaction with proper isolation handling. [\[Hard\]](#)

Recommended books

- Database System Concepts — Silberschatz, Korth, Sudarshan (McGraw-Hill) [Paid](#)
- Fundamentals of Database Systems — Elmasri & Navathe (Pearson) [Paid](#)
- DBMS (NPTEL) — IIT Madras (NPTEL) [Free](#) · nptel.ac.in

Additional resources

- ER + normalization notes
- SQL practice set
- Previous-year papers (2022-24)
- Lab manual (MySQL)

CS302 — Operating Systems (4 credits)

Processes, threads, scheduling, synchronization, deadlocks, memory management, file systems.

Assignments

1. Simulate FCFS and Round Robin CPU scheduling. [\[Easy\]](#)
2. Implement the producer-consumer problem using semaphores. [\[Medium\]](#)
3. Implement a page-replacement simulator (FIFO, LRU, Optimal). [\[Hard\]](#)

Recommended books

- Operating System Concepts — Silberschatz, Galvin, Gagne (Wiley) [Paid](#)
- OSTEP — Operating Systems: Three Easy Pieces — Arpaci-Dusseau (Open) [Free](#) · ostep.org

Additional resources

- OS notes
- Scheduling worksheet
- Previous-year paper (2024)

Master of Computer Applications (MCA)

Program code: MCA

Universities offering this program

- IGNOU
- NIT Warangal
- Pune University
- Jamia Millia Islamia
- VIT
- Christ University

Semester 1

MCA-101 — Advanced Data Structures & Algorithms (4 credits)

Balanced trees, hashing, graph algorithms, greedy & dynamic programming, NP-completeness intro.

Assignments

1. Implement an AVL tree with rotations. [\[Medium\]](#)
2. Solve the 0/1 knapsack using dynamic programming. [\[Medium\]](#)
3. Implement and analyze Kruskal's & Prim's MST. [\[Hard\]](#)

Recommended books

- Introduction to Algorithms (CLRS) — Cormen et al. (MIT Press) [Paid](#)
- Algorithms — Jeff Erickson (Open) [Free](#) · jeffe.cs.illinois.edu
- Algorithms (MIT OCW 6.006) — MIT (MIT OCW) [Free](#) · ocw.mit.edu

Additional resources

- Algorithm notes
- DP problem set
- Project ideas (10)
- Previous-year paper

Master of Business Administration (MBA)

Program code: MBA

Universities offering this program

- IIM Ahmedabad
- IIM Bangalore
- XLRI
- FMS Delhi
- SP Jain
- Symbiosis (SIBM)
- NMIMS

Semester 1

MBA-101 — Marketing Management (3 credits)

Marketing concepts, STP, the 4 Ps, consumer behaviour, branding and digital marketing fundamentals.

Assignments

1. Prepare an STP analysis for a consumer brand of your choice. [Easy]
2. Build a marketing mix (4 Ps) plan for a new product launch. [Medium]
3. Design a 90-day digital marketing campaign with KPIs and budget. [Hard]

Recommended books

- Marketing Management — Kotler & Keller (Pearson) Paid
- Principles of Marketing (OpenStax) — OpenStax (OpenStax) Free · openstax.org

Additional resources

- Case studies (5)
- Lecture notes
- Sample marketing plan template

MBA-102 — Financial Accounting (3 credits)

Accounting principles, journal & ledger, trial balance, final accounts, ratio analysis.

Assignments

1. Prepare journal entries and a trial balance from given transactions. [Easy]
2. Prepare final accounts (P&L + balance sheet) with adjustments. [Medium]
3. Perform ratio analysis and interpret a company's financial health. [Hard]

Recommended books

- Financial Accounting — Narayanaswamy (PHI) Paid
- Principles of Accounting (OpenStax) — OpenStax (OpenStax) Free · openstax.org

Additional resources

- Worked examples
- Practice problem set
- Previous-year paper

Bachelor of Business Administration (BBA)

Program code: BBA

Universities offering this program

- Christ University
- Symbiosis
- Amity University
- NMIMS
- Delhi University
- LPU

Semester 1

BBA-101 — Principles of Management (3 credits)

Functions of management — planning, organizing, staffing, leading and controlling; management thought.

Assignments

1. Explain the 14 principles of management with examples. [Easy]
2. Compare classical, behavioural and modern management theories. [Medium]
3. Analyze a real organization's structure and recommend improvements. [Hard]

Recommended books

- Principles of Management — Koontz & Weihrich (McGraw-Hill) Paid
- Principles of Management (OpenStax) — OpenStax (OpenStax) Free · openstax.org

Additional resources

- Lecture notes
- Case studies
- Previous-year paper (2024)

